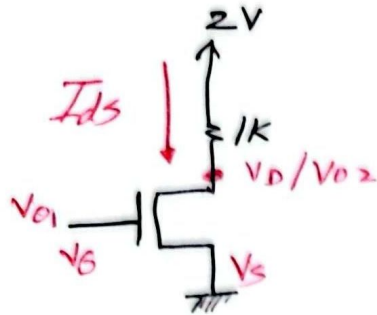


Problem 11



here,

$$V_T = 0.2$$

$$K_n = k_n' \frac{W}{L} = 4 \text{ mA/V}^2$$

$$V_{O1} = V_G = 2 \text{ V}$$

now, $V_{GS} = V_G - V_S = 2 - 0 = 2 > V_T$

$V_{GS} > V_T$ so not in cut off

assume saturation,

$$I_{DS} = \frac{K_n}{2} (V_{GS} - V_T)^2$$

$$\Rightarrow \frac{2 - V_d}{1k} = \frac{4}{2} (V_G - V_S - V_T)^2$$

$$\Rightarrow 2 - V_d = 2 (2 - 0 - 0.2)^2$$

$$\Rightarrow 2 - V_d = 2 \times 1.8^2$$

$$\Rightarrow 2 - V_d = 6.48$$

$$\therefore V_d = -4.48$$

$$V_{DS} = V_D - V_S = -4.48 - 0 = \boxed{-4.48} < \boxed{V_{GS} - V_T}$$

as negative.

V_{DS} neg so it's not in saturation.

Assume triode,

$$I_{ds} = \frac{kn}{2} [2(V_{GS} - V_T) V_{DS} - V_{DS}^2]$$

$$\Rightarrow \frac{2 - V_D}{1} = \frac{1}{2} [2(\underbrace{V_G}_{2} - \underbrace{V_S}_{0} - \underbrace{V_T}_{0.2})(V_D - V_S) - (V_D - V_S)^2]$$

$$\Rightarrow 2 - V_D = 2 [2(2 - 0.2) V_D - V_D^2]$$

$$\Rightarrow 2 - V_D = 2 [3.6 V_D - V_D^2]$$

$$\Rightarrow 2 - V_D = 7.2 V_D - 2 V_D^2$$

$$\Rightarrow 2 V_D^2 - 8.2 V_D + 2 = 0$$

$$\therefore V_D = 3.84, 0.26$$

now, $V_{GS} - V_T = V_{OV}$

$$V_{OV} = V_G - V_S - V_T$$

$$= 2 - 0 - 0.2$$

$$= 1.8$$

$$\boxed{V_{GS} > V_T}$$

$$2 > 0.2$$

for triode,

$$V_D = 3.84$$

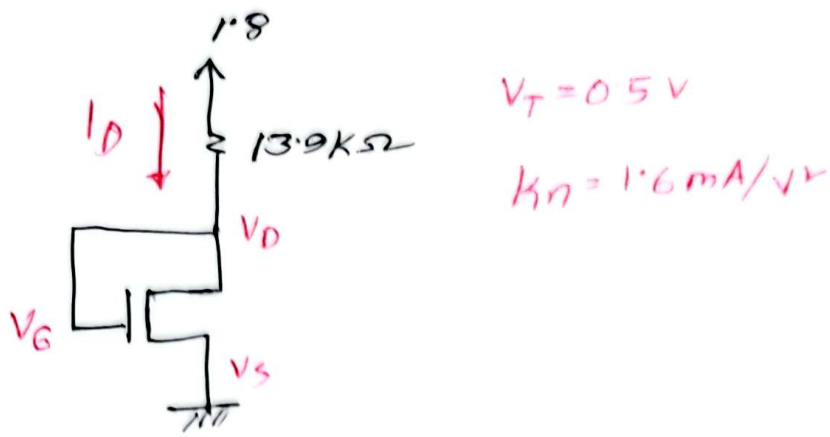
$$V_{DS} = V_D - V_S = 3.84 - 0 = 3.84$$

$$V_{DS} \neq (V_{GS} - V_T) / V_{OV} \text{ so not applicable}$$

but if, $V_D = 0.26$

$$\text{Then } V_{DS} = 0.26 - 0 = 0.26$$

$$\boxed{V_{DS} < V_{OV}}$$



here $V_D = V_G$ so mosfet is in sat or cutoff.

now if cutoff,

$$\begin{aligned}
 V_{GS} &= V_G - V_S \\
 &= 1.8 - 0 \\
 &= 1.8 > V_T
 \end{aligned}$$

so not cut off.

Saturation,

$$I_D = \frac{K_n}{2} [V_{GS} - V_T]^2$$

$$\Rightarrow \frac{1.8 - V_D}{13.9} = \frac{1.6}{2} [V_G - V_S - V_T]^2$$

$$\Rightarrow 1.8 - V_D = 13.9 \times 0.4 [V_D - 0 - 0.5]^2$$

$$\Rightarrow 1.8 - V_D = 11.12 [V_D - 0.5]^2$$

$$\Rightarrow 1.8 - V_D = 11.12 V_D^2 - 11.12 V_D + 2.78$$

$$\Rightarrow 11.12 V_D^2 - 10.12 V_D + 0.98 = 0$$

$$\therefore V_D = 0.79, 0.11$$

$$\text{for, } V_D = 0.79$$

$$\begin{aligned} V_{DS} &= V_D - V_S \\ &= 0.79 \end{aligned}$$

$$\begin{aligned} V_{OV} &= V_{GS} - V_T \\ &= 0.79 - 0 - 0.5 \\ &= 0.49 \end{aligned}$$

$$V_{DS} > V_{OV}$$

$$V_D = 0.11$$

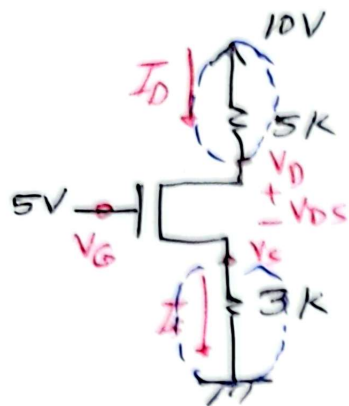
$$\begin{aligned} V_{DS} &= V_D - V_S \\ &= 0.11 \end{aligned}$$

$$V_{OV} = V_{GS} - V_T$$

$$\begin{aligned} V_{OV} &= 0.11 - 0 - 0.5 \\ &= -0.39 \end{aligned}$$

not possible

$\therefore$ so, $V_D = 0.79$ & MOSFET is in saturation.



$$V_T = 1$$

$$K_n = 2 \text{ mA/V}^2$$

If cut off,

$$I_D = I_S = I_{DS} = 0$$

$$V_D = 10 \quad V_G = 5 \quad V_S = 0$$

$$V_{GS} = 5 - 0 = 5$$

$$V_{GS} > V_T$$

so, not cut off.

here, $I_D = I_S = I_{DS}$

$$\Rightarrow \frac{10 - V_D}{5} = \frac{V_S - 0}{3}$$

assumption, sat,

$$I_{DS} = \frac{K_n}{2} [V_{GS} - V_T]^2$$

$$\Rightarrow \frac{V_S - 0}{3} = \frac{2}{2} [V_G - V_S - V_T]^2$$

$$\Rightarrow V_S = 3 [5 - V_S - 1]^2$$

$$\Rightarrow V_S = 3 [4 - V_S]^2$$

$$\Rightarrow V_S = 48 - 24V_S + 3V_S^2$$

$$\Rightarrow 3V_S^2 - 25V_S + 48 = 0$$

$$V_S = 5.333, 3$$

now, $V_s = 5.333$

$$V_{GS} = 5 - 5.333 \\ = -0.333$$

$V_{GS} < V_T$ not acceptable

as $V_{GS} > V_T$ for saturation

$$V_s = 3$$

$$V_{GS} = 5 - 3 = 2$$

$$\boxed{V_{GS} > V_T}$$

$$V_{OV} = V_{GS} - V_T \\ = 2 - 1 \\ = 1$$

$$V_{DS} = V_D - V_s \\ = 5 - 3 \\ = 2$$

$$\boxed{V_{DS} > V_{OV}}$$

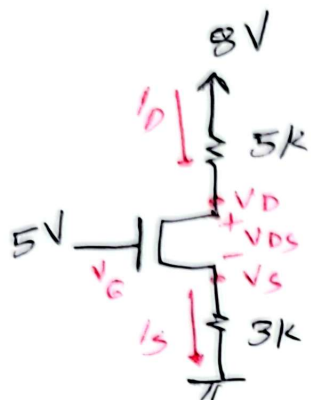
$$\cancel{V_D} \Rightarrow \frac{10 - V_D}{5} = \frac{V_s - 0}{3}$$

$$\Rightarrow \frac{10 - V_D}{5} = \frac{3}{3}$$

$$\Rightarrow 10 - V_D = 5$$

$$\Rightarrow V_D = 5$$

MosFET is in saturation.



$$V_T = 1$$

$$K_n = 2 \text{ mA/V}^2$$

If cutoff

$$I_D = I_S = I_{DS} = 0$$

$$V_G = 5$$

$$V_D = 8$$

$$V_S = 0$$

$$V_{GS} = 5 - 0 = 5 > V_T$$

like the previous one,

$$I_D = I_S$$

$$\Rightarrow \frac{8 - V_D}{5} = \frac{V_S - 0}{3}$$

$$\Rightarrow 8 - V_D = \frac{5}{3} V_S$$

saturation,

$$I_{DS} = \frac{K_n}{2} [V_{GS} - V_T]^2$$

$$\Rightarrow \frac{V_S - 0}{3} = \frac{2}{2} [V_G - V_S - V_T]^2$$

$$\Rightarrow \frac{V_S}{3} = [5 - V_S - 1]^2$$

$$\Rightarrow 3V_S^2 - 25V_S + 48 = 0$$

$$V_s = 5.333, 3$$

now, for $V_s = 5.333$,

$$V_{GS} = 5 - 5.333 \\ = -0.333$$

$$V_{GS} < V_T$$

not possible

$$V_s = 3$$

$$V_{GS} = 5 - 3 \\ = 2$$

$$V_{GS} > V_T$$

$$V_{OV} = V_{GS} - V_T \\ = 2 - 1 = 1$$

$$\Rightarrow 8 - V_D = \frac{5}{3} \cdot 3$$

$$\Rightarrow 8 - V_D = 5$$

$$\Rightarrow V_D = 3$$

$$V_{DS} = 3 - 3 \\ = 0$$

$$V_{DS} < V_{OV}$$

Not possible to be in sat.

assumption, ~~is~~ triode,

$$I_{Ds} = \frac{K_n}{2} [2(V_{GS} - V_T)(V_{DS}) - V_{DS}^2]$$

$$\Rightarrow \frac{8 - V_D}{5} = \frac{V_s}{3}$$

$$\Rightarrow 8 - \frac{5}{3} V_s = V_D$$

$$V_{DS} = 8 - \frac{5}{3} V_s - V_s$$

$$= \frac{24 - 5V_s - 3V_s}{3}$$

$$= \frac{24 - 8V_s}{3}$$

$$\Rightarrow I_{ds} = \frac{2}{2} \left[2(5 - V_s - 1) \frac{24 - 8V_s}{3} - \left(\frac{24 - 8V_s}{3} \right)^2 \right]$$

$$\Rightarrow \frac{V_s - 0}{3} = \left[\frac{(8 - 2V_s)(24 - 8V_s)}{3} - \frac{(24 - 8V_s)^2}{9} \right]$$

$$\Rightarrow 3V_s = (24 - 6V_s)(24 - 8V_s) - (24 - 8V_s)^2$$

$$\Rightarrow 3V_s = \cancel{24^2} - 8 \times 24V_s - 6 \times 24V_s + 48V_s^2 - 64V_s^2 + 384V_s - \cancel{24^2}$$

$$\Rightarrow 3V_s = -16V_s^2 + 48V_s$$

$$\Rightarrow 16V_s^2 - 45V_s = 0$$

$$V_s = 2.81, 0$$

$V_s \neq 0$, cause then $I_s = \frac{V_s - 0}{3} = \frac{0 - 0}{3} = 0$
not possible.

$$V_s = 2.81$$

$$V_{GS} = 5 - 2.81$$

$$= 2.19$$

$$V_{GS} > V_T$$

$$V_{OV} = 2.19 - 1$$

$$= 1.19$$

$$V_{DS} = V_D - V_s$$

$$= 8 - \frac{5}{3} V_s - V_s$$

$$= 0.506$$

$$V_{DS} < V_{OV}$$

active proved.